

SIMATS ENGINEERING

ASSESSMENT TOOL 2: LEVEL EXAM 2

COURSE NAME: Programming in Java

COURSE CODE: CSA0915- SLOT B

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EXAM QUESTIONS

1. Write a Program to Find the Nth Largest Number in a array.

Sample Input:

List : {14, 67, 48, 23, 5, 62}

N = 4

Sample Output:

4th Largest number: 23

OUTPUT:

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Enter the size of array:
Enter 6 elements:
N =
4th Largest number: 23

2. Write a program to count all the prime and composite numbers entered by the user.

Sample Input:

Enter the numbers

4

54

29

71

7

59

98

23

Sample Output:

Composite number:3

Prime number:5

OUTPUT:

The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the lab name, and a user profile for LOCHANA SRI R S. The main area is divided into 'Your Code' and 'Output' sections.

Your Code:

```
1 import java.util.*;
2
3 class Q2
4 {
5     public static void main(String[] args)
6     {
7         Scanner sc = new Scanner(System.in);
8
9         int p_count = 0;
10        int C_count = 0;
11
12        System.out.println("Enter the no. of integers to be entered");
13        int n = sc.nextInt();
14
15        System.out.println("Enter the numbers:");
16        for (int i = 0; i < n; i++)
17        {
18            int num = sc.nextInt();
19
20            if (num <= 1)
21            {
22                continue;
23            }
24
25            int factorCount = 0;
26            for (int j = 1; j <= num; j++)
27            {
28                if (num % j == 0)
29                {
30                    factorCount++;
31                }
32            }
33
34            if (factorCount == 2)
35            {
```

Output:

```
Enter the no. of integers to be entered:
Enter the numbers:
Composite number:3
Prime number:5
```

```

31         }
32     }
33
34     if (factorCount == 2)
35     {
36         p_count++;
37     }
38     else
39     {
40         C_count++;
41     }
42 }
43
44 System.out.println("Composite number:" + C_count);
45 System.out.println("Prime number:" + p_count);
46
47 }
48 }
49

```

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- Write a program to print the total amount available in the ATM machine with the conditions applied.

Total denominations are 2000, 500, 200, 100, get the denomination priority from the user and the total number of notes from the user to display the total available balance to the user

Sample Input:

Enter the 1st Denomination: 500
Enter the 1st Denomination number of notes: 4
Enter the 2nd Denomination: 100
Enter the 2nd Denomination number of notes: 20
Enter the 3rd Denomination: 200
Enter the 3rd Denomination number of notes: 32
Enter the 4th Denomination: 2000
Enter the 4th Denomination number of notes: 1

Sample Output:

Total Available Balance in ATM: 12400

OUTPUT:

Your Code

```
1 import java.util.*;  
2  
3 class Q3  
4 {  
5     public static void main(String[] args)  
6     {  
7         Scanner sc = new Scanner(System.in);  
8         int balance = 0;  
9  
10        System.out.println("Enter the number of denominations: ");  
11        int n = sc.nextInt();  
12  
13        for (int i = 0; i < n; i++)  
14        {  
15            System.out.println("Enter the denomination: ");  
16            int denomination = sc.nextInt();  
17  
18            System.out.println("Enter the denomination no. of notes");  
19            int notes = sc.nextInt();  
20  
21            balance += denomination * notes;  
22        }  
23  
24        System.out.println("The Account Balance is " + balance);  
25  
26    }  
27 }  
28
```

Output

```
Enter the number of denominations:  
Enter the denomination:  
Enter the denomination no. of notes:  
Enter the denomination:  
Enter the denomination no. of notes:  
Enter the denomination:
```

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4. Write a program to read a character until a * is encountered. Also count the number of uppercase, lowercase, and numbers entered by the users.

Sample Input:

Enter * to exit...

Enter any character: W

Enter any character: d

Enter any character: A

Enter any character: G

Enter any character: g

Enter any character: H

Enter any character: *

Sample Output:

Total count of lower case:2

Total count of upper case:4

Total count of numbers =0

OUTPUT:

Your Code

```
1 import java.util.*;
2 class Q4
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         System.out.println("Enter * to exit");
8         int lowercase =0;
9         int Uppercase =0;
10        int numbers = 0;
11        while(sc.hasNext())
12        {
13
14            char Ch = sc.next().charAt(0);
15
16            if( Ch =='*')
17            {
18                break;
19            }
20
21            if(Ch>='a' && Ch <='z')
22            {
23                lowercase++;
24            }
25            else if(Ch>='A' && Ch<='Z')
26            {
27                Uppercase++;
28            }
29            else if(Ch>='0' && Ch<='9')
30            {
31                numbers++;
32            }
33        }
34    }
```

Output

```
Enter * to exit
Lower cases: 3
Upper Cases: 7
Numbers: 2
```